ANX-PR/CL/001-01
LEARNING GUIDE

SUBJECT

45000127 - Surface Engineering

DEGREE PROGRAMME

04MI - Grado En Ingenieria De Materiales

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 2

Index

Learning guide

1. Description

1.1. Subject details

Name of the subject	45000127 - Surface Engineering
No of credits	6 ECTS
Type	Compulsory
Academic year ot the programme	Third year
Semester of tuition	Semester 6
Tuition period	February-June
Tuition languages	English
Degree programme	04MI - Grado en Ingenieria de Materiales
Centre	04 - E.T.S. De Ing. De Caminos Canales Y P.
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Miguel Angel Sanchez Garcia (Subject coordinator)	B-107	miguelangel.sanchez@upm.es	Sin horario. These will be announced at the beginning of the semester.
Alberto Bosca Mojena	B-308	alberto.bosca@upm.es	Sin horario. These will be announced at the beginning of the semester

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Conocimientos básicos sobre física moderna. Inglés, tanto ecrito como oral.

4. Skills and learning outcomes *

4.1. Skills to be learned

CE 2. - Saber modelizar el comportamiento (mecánico, electrónico, químico o biológico) de los materiales y su integración en componentes y dispositivos.

CE 6. - Saber diseÃ±ar, evaluar, seleccionar y fabricar materiales segÃºn sus aplicaciones

CG 11 - Responsabilidad y ética profesional

CG 3 - Comunicación oral y escrita

CG 9 - Capacidad de trabajo interdisciplinar

4.2. Learning outcomes

RA13 - Saber adaptarse a nuevas situaciones, ejecutando el trabajo con responsabilidad y respeto a los demás.

RA11 - Saber utilizar técnicas para modificar las superficies e intercaras para influir en el comportamiento deseado del material

RA10 - Conocer el comportamiento físico-químico de las superficies e intercaras y su influencia en las propiedades mecánicas, electrónicas, químicas y biológicas

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

The main objective of the course is for the students to achieve a basic knowledge on different technological processes that are applied to materials used mainly in the nano and microelectronic fields. Furthermore, the course will also analyze the main effects that those technological processes have on the optical and electrical properties of different optoelectronic devices.

5.2. Syllabus

1. Properties of Semiconductor Materials
    1.1. Classification of materials
    1.2. Crystalline Structures
    1.3. Impurities (doping) in semiconductors
    1.4. Crystal Defects in semiconductors
2. Semiconductor Fabrication Techniques
    2.1. Fabrication of pure materials
    2.2. Czochralsky growth
    2.3. Float Zone
    2.4. Bridgman Technique
3. Epitaxial Techniques

3.1. Liquid Phase Epitaxy (LPE)

3.2. Molecular Beam Epitaxy (MBE)

3.3. Chemical Vapor Deposition (CVD)

4. Doping Techniques

4.1. Diffusion

4.2. Ion Implantation

5. Fabrication of non-semiconductor materials

5.1. Thermal oxidation (dry and wet)

5.2. Chemical vapor deposition applied to the fabrication of dielectric materials

6. Lithography Techniques

6.1. Standard optical lithography

6.2. Nanolithography (e-beam)

7. Metallization

7.1. Joule effect

7.2. Electron Beam technique

7.3. Sputtering technique

8. Etching, bonding and encapsulating

8.1. Dry and chemical wet etching

8.2. Wire bonding, flip-chip bonding and tape-automated bonding

8.3. Packaging

9. Integrated devices and state of the art

9.1. Passives and MOSFETs

9.2. Industry challenges: High and low k dielectrics, FinFET technology, interconnects

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	**Tema 1. Properties of Semiconductor Materials: Classification of materials, Crystalline Structures, Doping** Duration: 04:00 Lecture			
2	**Tema 2. Semiconductor Fabrication Techniques: Fabrication of pure materials** Duration: 04:00 Lecture			
3	**Tema 2 (cont). Semiconductor Fabrication Techniques: Czochralsky growth, Float Zone and Bridgman Technique** Duration: 04:00 Lecture			
4	**Tema 3: Epitaxial Techniques: Liquid Phase Epitaxy (LPE)** Duration: 04:00 Lecture			
5	**Tema 3 (cont). Epitaxial Techniques: Molecular Beam Epitaxy (MBE) and Chemical Vapor Deposition (CVD)** Duration: 04:00 Lecture			
6	**Review and Problems Section (Chapters 1 to 3)** Duration: 04:00 Lecture			**Examen** Written test Progressive assessment Presential Duration: 00:00
7	**Tema 4. Doping Techniques: Diffusion** Duration: 04:00 Lecture			
8	**Tema 4 (cont). Doping Techniques: Ion Implantation** Duration: 04:00 Lecture			
9	**Tema 5. Fabrication of non-semiconductor materials: Thermal oxidation (dry and wet)** Duration: 04:00 Lecture			

10	**Tema 5 (cont). Fabrication of non-semiconductor materials: Chemical vapor deposition applied to the fabrication of dielectric materials** Duration: 04:00 Lecture			
11	**Tema 6. Lithography Techniques: Standard optical lithography and Nanolithography (e-beam).** Duration: 04:00 Lecture			
12	**Tema 7. Metallization: Joule effect.** Duration: 04:00 Lecture			
13	**Tema 7 (cont) .Metallization: Electron Beam technique and Sputtering technique** Duration: 04:00 Lecture			
14	**Tema 8. Etching, bonding and encapsulating: Dry and chemical wet etching** Duration: 04:00 Lecture			
15	**Tema 9. Integrated devices and state of the art** Duration: 04:00 Lecture			
16				
17				**Examen** Written test Progressive assessment Presential Duration: 03:00 **Examen final** Written test Global examination Presential Duration: 03:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
6	Examen	Written test	Face-to-face	00:00	50%	5 / 10	CE 6. CG 3 CG 9 CG 11 CE 2.
17	Examen	Written test	Face-to-face	03:00	50%	5 / 10	CE 2. CE 6. CG 3 CG 9 CG 11

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Examen final	Written test	Face-to-face	03:00	100%	5 / 10	CE 2. CE 6. CG 3 CG 9 CG 11

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Examen Final	Written test	Face-to-face	03:00	100%	5 / 10	CE 2. CE 6. CG 3 CG 9 CG 11

7.2. Assessment criteria

The final grade of the course will be calculated in the following ways:

PROGRESSIVE EVALUATION: Final grade = 50% of the first partial exam + 50% of the second partial exam. The first partial exam will be held around week 7 of the course and it will cover the material covered in class until that date. This exam will have a weight of 50% of the final grade and there is a minimum grade of 5.0 points in order to be considered. The other 50% will come from a second partial exam covering the rest of the program of the course and that will be held during the period of Final Exams. This second exam will also have a minimum grade of 5.0 points in order to be considered. For those students who did not get the minimum grade in the first partial exam, they will perform a final full exam with 100% weight for the final grade.

EVALUATION based on a FINAL exam (GLOBAL): The final grade will be that obtained in a single written final exam covering all the chapters of the course.

EXTRAORDINARY EVALUATION: The final grade of the Extraordinary Evaluation will be that obtained in a single written final exam covering all the chapters of the course.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
S.M. Sze, "Physics of Semiconductor Devices". John Willey & Sons, 3rd edition (2007)	Bibliography	Reference book
G.S. May, S.M. Sze, "Fundamentals of Semiconductor Fabrication", John Wiley & Sons (2003)	Bibliography	Reference book
P. Van Zant, Microchip fabrication: a practical guide to semiconductor processing, Sixth edition. New York: McGraw-Hill Professional, 2014	Bibliography	Reference book

Virtual Classroom	Web resource	Moodle platform of the course

9. Other information

9.1. Other information about the subject

All the additional documentation to follow this course will be available at the Moodle platform of the course.

In case there is a forced confinement situation due to COVID-19, all the lectures will be available in video files. The exams will be carried out using the Moodle platform of the course.